



Society of Petroleum Engineers

SPE-229760-MS

Navigating the Challenge to Recycle Large Ships and the Role of Various Stakeholders in Advancing Responsible Recycling

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This paper was prepared for presentation at the ADIPEC held in Abu Dhabi, UAE, 3 – 6 November 2025.

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Abstract

The global shipping industry is at a critical stage as it faces an unprecedented challenge in responsibly recycling large vessels exceeding 50,000 gross tonnages (GT). These huge ships, including post-Panamax, Suezmax, & ultra-large container vessels, along with Very Large Crude Carriers (VLCCs). Ultra Large Crude Carriers (ULCCs), and Ultra Large Gas Carriers (ULGCs) represent an integral part of the international maritime trade yet pose unique complexities when they reach the end-of-useful-life phase. One of the biggest challenges in front of the industry is that there aren't enough recycling facilities that can handle these huge ships safely and responsibly. Some of the other impending issues related to the recycling of these large vessels have been discussed in detail in this paper, including the regulatory landscape, infrastructural limitations (limited supply), environmental considerations, and economic feasibility.

Large vessels require specialised shipbreaking facilities that demand substantial capital investment, mainly to accommodate such ships. Few critical infrastructural investments include dredging costs to comply with adequate draft depths of 8-10 meters, berth or drydock width of over 40 meters, and heavy-lift cranes capable of handling over 50-ton blocks. The challenge is further amplified considering that approximately 16,000 vessels (representing over 700 million deadweight tonnes - DWT) are expected to reach their end-of-life and require recycling over the next decade. This projection is nearly three times the ships recycled over the last 10 years. (MI News Network 2025)

This paper evaluates the major challenges regarding the recycling of large ships, assessing the regulatory landscape with special emphasis on the IMO's Hong Kong International Convention (HKC) ratification, market dynamics, infrastructure constraints, and industry stakeholder responsibility. The analysis addresses the urgent need for green ship recycling that balances environmental protection, worker safety, and financial viability while meeting the growing demand for responsible ship recycling. Some emphasis has also been placed on the underlying opportunity for a circular economy and the potential supply of green steel.

Introduction and Context

The global shipping industry is at a critical stage as it faces an unprecedented challenge in responsibly recycling large vessels exceeding 50,000 gross tonnages (GT). These huge ships, including post-Panamax, Suezmax, & ultra-large container vessels, along with Very Large Crude Carriers (VLCCs). Ultra Large

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Understanding the demand-supply scenario

With the most optimistic assumptions regarding operational efficiency, recycling shipyards with around 200,000 LDT annual throughput (average capacity for most upcoming facilities) can only cater up to 5-6 large vessels (~30,000 LDT) annually. It is estimated that currently, less than 30 yards globally are capable of handling 30,000 LDT vessels globally, of which the ones compliant with the Hong Kong International Convention for the Safe and Environmentally Sound Recycling of Ships (HKC) and EU Ship Recycling Regulations (EU SRR) are relatively scarce in number. As of February 2025, there are only 13 facilities on the EU list that are capable of handling post-Panamax-sized vessels - 4 shipyards within the EU, 8 shipyards in Turkey, and 1 shipyard in the US.

The global shipping industry is likely to experience a record surge in recycling demand, with BIMCO projecting that around 16,000 ships (representing around 700 million DWT) will require recycling over the next decade. This represents a dramatic increase from the previous decade, during which 7,780 ships totalling 285 million DWT were recycled, indicating a 3X tonnage volume (MI News Network 2025).

The surge is primarily driven by the age profile of the current fleet and regulatory pressures. Ships built during the 2000's construction boom, when tonnage increased by almost 115% and the number of vessels grew by ~31%, are likely approaching their typical 25-30 year operational lifespan. This fleet represents a significantly larger volume than the 1990s-built vessels that dominated scrapping activity between 2015-2024 (Safety4Sea 2023). The timing coincides with the enforcement of several stringent environmental regulations, including the Carbon Intensity Indicator (CII) and IMO's greenhouse gas emissions targets, which are accelerating retirement decisions for older, less efficient vessels (Clarksons Research 2023) (Splash247 n.d.).

As per our estimates, over 1,300 vessels (over 30,000 LDT) are to be sent for scrapping between 2024 to 2037 (a number which could easily go up to 1,500 - 1,600 vessels with stringent regulatory push and suitable market conditions), around 85% (or 1,110+ vessels) are non-OECD flagged, while 189 of them are EU-flagged ships. Further, approximately 4,000+ vessels (with size >30,000 LDT) across containerships, tankers, bulk carriers, and gas carrier segments are expected to be rated D-E under CII regulations as per their carbon emissions intensity.

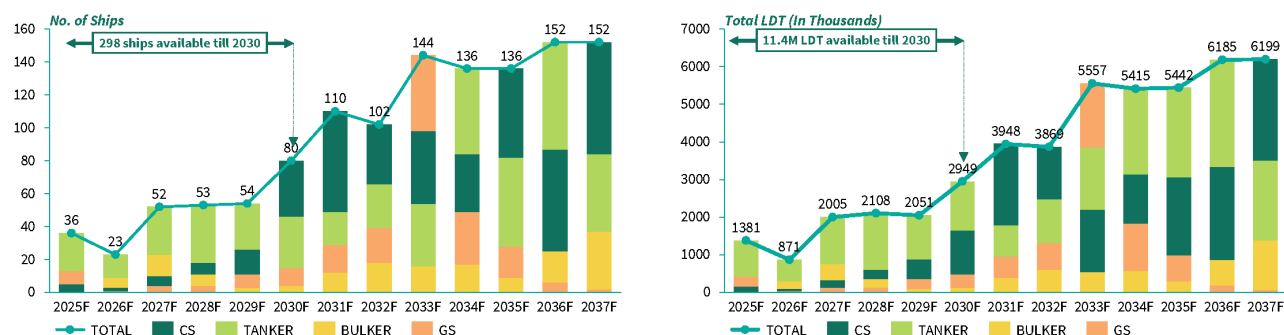


Figure 1—Number of vessels available for ship recycling (ship size ≥ 30k LDT), Clarksons Research

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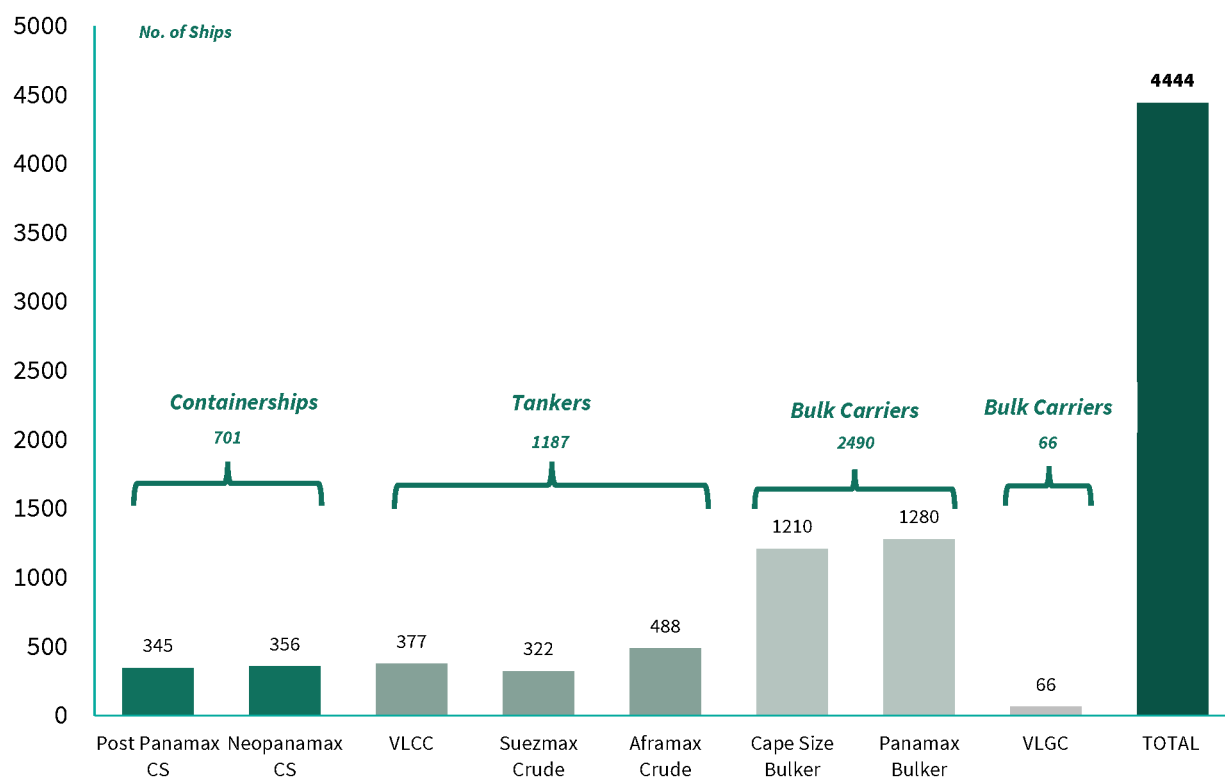


Figure 2—Number of vessels expected to be rated D-E in 2026

This indicates a huge capacity deficit to cater to the expected influx of large vessels reaching their end-of-useful-life over the next decade. Especially, there is a noteworthy gap between the current and the required capacity to recycle the EU-flagged large vessels. Urgent action is needed to ensure that the growth in demand for ship recycling services is met by facilities with responsible practices.

Market Dynamics Keep Ships at Sea Longer

Market cyclicality currently presents a complex situation where strong freight rates and operational demand are keeping older vessels in service longer than normal, temporarily suppressing recycling activity (Safety4Sea 2024). The Baltic Dry Index (BDI) and container freight indices remained elevated through 2022 - 2024, making even the aging vessels profitable to operate (Shipping Box 2025). The ship recycling

market has also experienced certain imbalance due to the ongoing geopolitical turmoil in the Middle East, Red Sea disruptions, and Panama Canal diversions, which have extended voyage times and increased vessel utilization ([Shipping Box 2025](#)). This has created a supply-demand imbalance where recycling volumes have declined significantly from peak levels, with some estimates even showing 50% reduction in dismantling volumes between 2021 and 2023 ([Safety4Sea 2024](#)).

However, this current market depression is expected to reverse dramatically as freight rates stabilize, and the accumulated volume of aging vessels becomes economically unviable. The challenge is particularly acute for large vessels, where the capacity constraints are most severe. Post-Panamax and Suezmax vessels face unique challenges due to their size requirements, with many existing facilities unable to accommodate vessels exceeding standard dimensions ([MFAME 2024](#)).

The regulatory push from HKC implementation adds urgency to this demand surge. Major shipping companies like Maersk have publicly acknowledged challenges in finding compliant facilities capable of handling post-Panamax vessels responsibly ([MFAME 2024](#)). This has prompted exploration of alternative solutions, for instance, the development of green field facilities on India's East Coast and upgraded capabilities in existing hubs (P Manoj; [ET Infra 2024](#)).

Regulatory Landscape and HKC Ratification

The enforcement and ratification of the IMI's 2009 Hong Kong International Convention (HKC) for the Safe and Environmentally Sound Recycling of Ships on 26 June 2025 marks a defining moment in the global ship recycling regulations ([IMO 2025](#)). The HKC, after 16 years since its adoption in 2009, finally became a law. 24 nations have now ratified the Convention, representing 57% of the global shipping tonnage ([IMO 2025](#)). The HKC establishes binding international standards that will transform how ships are recycled globally.

The Convention applies to all ships equal to or more than 500 GT engaged in international trade, requiring comprehensive compliance from both shipping companies and ship-breaking facilities. It requires that every ship must maintain an Inventory of Hazardous Materials throughout its working life ([IMO 2025](#)). Furthermore, ships can only be broken down at facilities with appropriate authorization from government authorities. All HKC-compliant shipyards must secure a Statement of Compliance (SoC) from a competent third-party company, typically a classification society. Each vessel requires a specific ship recycling plan (SRP) before it can be broken up. ([IMO 2025](#))

For large ships, these stringent norms create multi-fold challenges. To handle such large ships, more than 50,000 GT (<12, 000 Light Displacement Tonnage - LDT, as generally used in ship recycling), facilities must have adequate draft depths - at least 8- 10 meters draft, 40-70 meters of width to handle post-Panamax vessel class, cranes must be capable to lift at least 50-ton blocks, and a safe waste handling (and disposal) systems ([MFAME 2024](#)). As of July 2025, most of the existing shipyards fall short of these requirements.

As of February 2024, only 111 Indian yards (according to a publication of Rajya Sabha Committee on Transport, Tourism and Culture), four ship recycling yards in Bangladesh, and one Canadian yard has been provided Statement of Compliance (SoC) by international classification societies as compliant with the HKC standards. Further, as the Hong Kong Convention (HKC) for Ship Recycling is set to be enforced in 2025, the industry is at a pivotal juncture, necessitating a paradigm shift in recycling practices. The ratification of HKC adds complexity, requiring global adherence to environmentally sound standards.

The European Union Ship Recycling Regulations (EU-SRR) add another layer of complexity, setting even higher standards than the HKC (8). The EU-SRR bans beaching methods completely and requires stricter downstream waste management. EU Member State- flagged ships can only be recycled at a facility on the EU List. Of the 43 shipyards recognized in the EU List, 4 shipyards in 2024 and 12 shipyards in 2025 were due for renewal of their inclusion in the list. Among the current yards in the EU-list, only around 13 shipyards are capable of handling large vessels (>30,000 LDT) (Basis of the information of draft, length,

and width of the facility as per the EU- list), leaving a huge capacity gap for large vessels (P Manoj; [ET Infra 2024](#)).

Since 2016, 27 Indian ship recycling facilities have applied to the EU List, but none have been accepted. With regards to Bangladesh and Pakistan, no facilities have applied for inclusion on the EU List as per the last update on 08 December 2023. As of February 2025, only 14 non-EU yards have made it to Part B of the EU List of recycling facilities and 11 yards have completed step 2 of the application process for inclusion on the EU List. Of the 14 non-EU yards on the Part B, 11 are from Turkiye. (P Manoj; [ET Infra 2024](#))

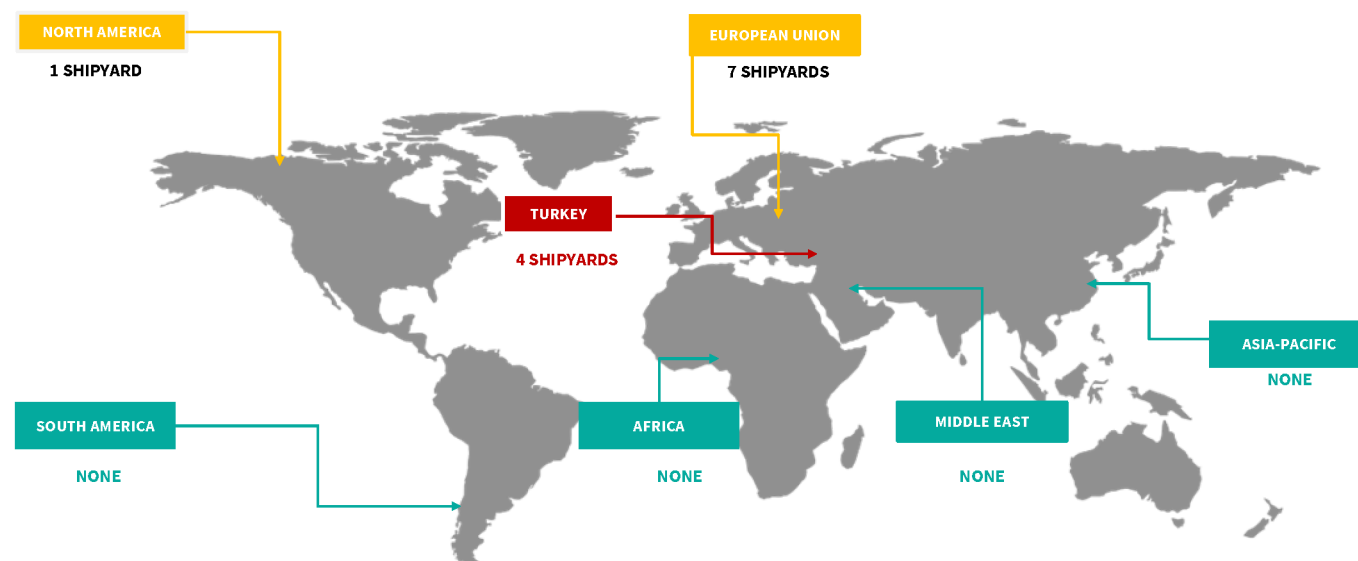


Figure 3—EU List Post Panamax recycling capacity (On Paper)

The inclusion of non-OECD country yards on the EU List is crucial to overcoming capacity constraints. However, aligning existing yards with the HKC standards and fostering responsible recycling practices becomes imperative for sustainable and compliant recycling of all vessel sizes, including the post-Panamax and Suezmax vessels.

In addition to these widely accepted regulations, many countries have introduced/intend to introduce stricter norms in ship recycling to move towards their own wider sustainability goals. As an example, while neither EU SSR nor HKC regulations cover the landing of the vessels before recycling in their guidelines, the United Arab Emirates (UAE) (vide Circular No. (19) of 2023) prohibits landing/beaching of vessels in UAE shipyards. This mandates the construction of a jetty, further complicating regulations and increasing capital expenditure. ([Mulinaris 2023](#))

Further, the Basel Ban Amendment which prohibits the export of hazardous waste from OECD (Organization for Economic Cooperation & Development) countries to developing countries. Since ship recycling involves various quantities of hazardous waste such as asbestos, mercury, Chlorofluorocarbons (CFCs), POPs, etc, the business is impacted by the restrictions placed by the act. There might be restrictions on the purchase of OECD country-flagged vessels by recycling facilities in the non-OECD nations. ([International Pollutants Elimination Network \(IPEN\) 2020](#))

Given the urgency of this challenge, it is crucial to build or upgrade existing facilities to green ship recycling yards capable of recycling these large vessels in compliance with HKC/EU SRR regulations in alignment with international industry best practices.

The Infrastructure Reality Check

As already established in previous sections, large ships require specialized infrastructure that most of the current yards merely have. The technical requirements go far beyond just having a big beach or dry dock.

Post-Panamax vessels require a minimum draft depth of 8-10 meters to be positioned safely. Many traditional beaching operations can't accommodate these depths. Ships must be positioned during high tides and accessed across tidal zones, creating operational challenges and safety risks.

Berth width and LOA present another constraint. Post-Panamax ships need at least 40-70 meters of clear width and a minimum length of 170 m for safe operations. Traditional plots at facilities like Alang (India) are restricted to 150 meters, requiring expansion and amalgamation to handle larger vessels. The Gujarat Maritime Board is working on policy changes to create larger plots, but this will take time and hefty investment.

Heavy-lift cranes capable of handling 50+ tonne sections are essential for safe large ship recycling. Many existing facilities rely on smaller equipment designed for conventional vessels. Upgrading crane capacity will require significant capital investment and structural modifications.

Cutting equipment must also handle thicker steel sections found in large ships. Advanced mechanized cutting systems reduce safety risks compared to manual torch cutting; however, they require higher investment and skilled operators.

Case for Major Asian Ship Recycling Hubs - India, Bangladesh, and Pakistan

India's Alang facility demonstrates both the scale and constraints of the current recycling infrastructure (23). With 153 plots and 4.5 million LDT capacity, Alang handles over 98% of India's ship recycling ([Rai, Sinha and Murlidhar 2024](#)). The facility has recycled ships exceeding 80,000 tonnes LDT, proving its capability for large vessel operations (Thorat, Kiran; Noor, Nayeem; [GMS 2025](#)).

However, Alang faces operational challenges with the largest ships. Vessels exceeding plot capacity must be beached 500 meters away and manually pulled to recycling positions by workers. Major shipping companies like Maersk consider this practice incompatible with responsible recycling standards, limiting their willingness to use the facility for post-Panamax vessels (P Manoj; [ET Infra 2024](#)) ([MFAME 2024](#)).

The region employs approximately 30,000 workers directly in ship recycling operations. Most operational yards have implemented standard procedures aligned with international safety and environmental expectations, including impermeable flooring, effective drainage systems, and designated hazardous waste containment areas (Thorat, Kiran; Noor, Nayeem; [GMS 2025](#)).

Bangladesh's Chattogram district represents another major hub, but compliance remains patchy (Nicola Mulinaris; [NGO Shipbreaking Platform 2025](#)). Of approximately 50 operational yards, only four have achieved green ship recycling certification (Nicola Mulinaris; [NGO Shipbreaking Platform 2025](#)). The top facilities showcase impressive technological capabilities, but the majority operate below international (EU-SRR) standards (Marprof Environmental Ltd; [BIMCO 2024](#)).

Recent ship arrivals in Bangladesh show growing confidence in compliant facilities. Two LNG vessels - SUR and RASI - each over 30,000 LDT, were successfully recycled at Hong Kong Convention-aligned yards in July 2025 ([Shipping Box 2025](#)). This signals international recognition of Bangladesh's reform efforts, but capacity remains limited.

Pakistan's Gadani facility faced challenges with recurring safety incidents (Marprof Environmental Ltd; [BIMCO 2024](#)). A 2016 gas cylinder explosion killed 29 people, highlighting inadequate safety procedures (Nicola Mulinaris; [NGO Shipbreaking Platform 2025](#)). These incidents led to periodic closures that eliminated large-scale operations and damaged the facility's reputation for handling complex vessels safely.

Green Ship Recycling and the Circular Economy

Ship recycling offers huge environmental benefits when done right. The steel recovered from recycled ships produces only one-fifth the greenhouse gas emissions compared to making steel from iron ore (RL Kalthia Shipbreaking n.d.). Every tonne of ship scrap saves 1.3 tonnes of iron ore, 780 kg of coal, and 270 kg of limestone (RL Kalthia Shipbreaking n.d.).

The numbers are impressive. Since 1982, India's ship recycling industry has produced nearly 67 million tonnes of green steel (RL Kalthia Shipbreaking n.d.). That's equivalent to building 9,077 Eiffel Towers and saving 9 years of New York City's daily power consumption (RL Kalthia Shipbreaking n.d.).

Asian recycling facilities achieve 95-97% material recovery rates compared to 93% in European yards (RL Kalthia Shipbreaking n.d.) (Mishra Kajal; [Safety4Sea 2023](#)). Ships contain 80-85% steel, but also valuable non-ferrous metals, machinery, and equipment that find new applications. High-quality ship steel commands premium prices because of its fine-grain structure and superior corrosion resistance (Mishra Kajal; [Safety4Sea 2023](#)) (RL Kalthia Shipbreaking n.d.).

The circular economy model extends beyond steel recovery. Ship components get refurbished for marine applications. Equipment finds new uses in construction and heavy machinery. Even furniture and fittings get reused, supporting local economies while reducing demand for virgin materials (Mishra Kajal; [Safety4Sea 2023](#)).

Integration with Steel Industry Decarbonization

The steel industry accounts for 7-9% of global greenhouse gas emissions. India's steel sector represents 12% of the country's total emissions, presenting major decarbonization opportunities through increased scrap utilization ([Rai, Sinha and Murlidhar 2024](#)).

Currently, India's ship recycling contributes about 0.5% to total steel production. Climate Catalyst estimates that doubling India's recycling capacity by 2028 could avoid up to 20 million tonnes of emissions by 2040 ([Rai, Sinha and Murlidhar 2024](#)). This positions ship recycling as a key lever for steel industry decarbonization globally.

Electric arc furnaces (EAFs) using recycled steel require only one-third the energy needed for primary steel production (Hiremath, Dr. Anand M.; [GMS 2023](#)). As steel mills transition to cleaner production methods, demand for recycled ship steel will increase, improving recycling economics and supporting facility investment.

Corporate Sustainability Drives Demand

Major shipping companies increasingly prioritize certified green recycling over cost considerations (Maersk n.d.). Corporate sustainability commitments, investor expectations, and regulatory requirements drive demand for transparent, responsible recycling practices.

The green premium for sustainable recycling becomes economically viable as carbon pricing mechanisms create cost advantages for low-emission steel production ([Elmenshawy, Ulku and Hsuan 2024](#)). This convergence of environmental necessity and economic opportunity positions green ship recycling as essential for sustainable maritime industry transformation.

Addressing the urgency

The challenge of recycling large ships over 30,000 LDT represents one of the most complex issues facing the global maritime industry. The conjunction of increasing supply, regulatory requirements, and infrastructure limitations creates an urgent need for coordinated industry response to prevent bottlenecks that could compromise responsible recycling practices.

To address the above demand-supply gap to responsibly recycle larger vessels, the shipping industry (shipyards) can opt for either of the two options as per their suitability -

- A. Expansion and upgrade of existing brownfield shipyards as well as repair yards to meet the regulatory pre-requisites of the HKC and the EUSRR.
- B. Building greenfield integrated ship repair & recycling facilities capable of handling vessels of over 30,000 LDT size band.

Most of the existing facilities are established in major ship recycling hubs including India, Bangladesh, Pakistan, and Turkey. The existing hubs have well-established downstream markets and existing supply chains to handle the scrap metals (ferrous as well as nonferrous) and used ship components & machinery. The brownfield (existing) ship recycling facilities need to upgrade the facilities to align with HKC standards for authorisation. Alongside, the shipyards must build a proper business case for the planned recycling facilities.

A proper business case must provide a detailed financial model accounting for CAPEX, potential revenue streams from off take of scrap metals & used machinery/equipment, OPEX, manpower planning, taxation, expected cashflow for the next 15-20 years of operations, project's profitability, return on investment (ROI), and gauge project's financial attractiveness through IRR calculations. This detailed business case will provide a comprehensive understanding of the recycling facility's economic viability and assist in making informed decisions for project stakeholders.

With the projected increase in ship recycling volumes in the coming decade, it is going to be challenging to meet the demand for green ship recycling with the current setup. There is an urgent need to upgrade existing ship repair and recycling facilities to green ship recycling facilities and set up new ones that are capable of handling even larger ships such as the Suezmax, post-Panamax vessels and UL/VLCCs, among other categories.

Additionally, there lies a huge opportunity for repair shipyards globally that can be upgraded to offer ship recycling services to all types and sizes of vessels. As of December 2023, there were over 970 dedicated ship repair facilities globally.

However, it should be also noted that ship recycling is a cyclical business and depends on multiple factors. Hence, it will not be commercially viable to set-up stand-alone recycling shipyards. It will be more commercially lucrative to operate an integrated ship recycling, repair & maintenance, and ship building facility to solve the above issue. Integrated ship recycling facilities offer several advantages over stand-alone repair or recycling yards including improved capacity utilization rates, optimised OPEX, revenue gains, and lower CAPEX than greenfield facilities.

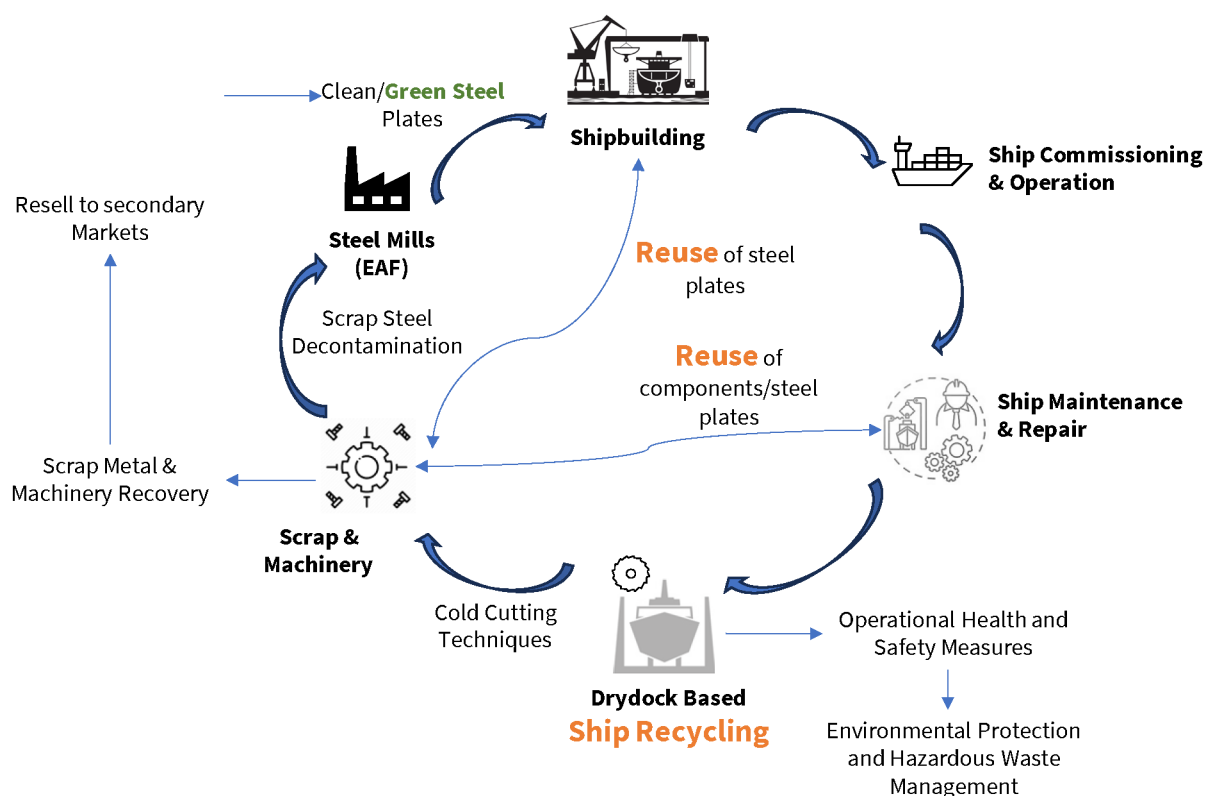


Figure 4—Typical role of green ship recycling in material circularity and green steel economy

Key actions up on the recycling shipyards include:

- **Authorisation process:** Request authorisation from competent authorities as per HKC guidelines.
- **SRFP and SRP development:** Develop and maintain a Ship Recycling Facility Plan (SRFP) and ship-specific Recycling Plans (SRP) for each vessel.
- **Compliance training:** Device training programs for safe and environmentally sound operations.
- **Emergency preparedness:** Establish and implement emergency readiness and response plans.
- **Infrastructure upgrades:** Make necessary improvements to the facilities to meet the latest safety and environmental standards.
- **Conduct a thorough upstream and downstream market assessment** by pre-empting the key ships reaching the EoL and tracking major ship owners' initiatives towards ship recycling & circularity in their overall sustainability motives. Also, support in establishing collaborations with the scrap dealers and steel mills (HRC facilities) for off-take.
- **Explore green steel and circularity opportunities:** Recycling shipyards and shipping companies must draw inspiration from industry leaders like Maersk, exploring opportunities for green steel production and integration with circular economy initiatives.
- **Develop a robust business plan:** a holistic financial business plan must be developed to check the viability of the facility (both operational and commercial) and will help attract stakeholders and shipping companies to get long-term benefits.

The Path Forward

The challenge of recycling large ships over 30,000 LDT represents one of the most complex issues facing the global maritime industry. The conjunction of regulatory changes, high influx of vessel indicating escalated recycling demand, and severe infrastructure constraints creates an urgent need for coordinated action.

The regulatory landscape, anchored by the recent implementation of the HKC globally along with the EU-SRR and other regional regulations related to responsible ship recycling, requires mandatory global standards that essentially govern ship-breaking practices. However, the limited capacity of compliant facilities to handle large vessels creates critical roadblocks.

The technical challenges extend beyond the simple scale considerations to include specialized infrastructure requirements, advanced environmental management systems, and operational capabilities that align with global best practices. Current facilities require significant upgrades to handle the large vessel segments responsibly while maintaining high environmental and safety compliance.

Green ship recycling emerges as both an environmental friendly practice and an economic opportunity. With recycled steel producing one-fifth of the emissions of conventional production and achieving excellent material recovery rates, the sector represents a crucial constituent of the maritime decarbonization strategies.

Successful scale-up of the sector requires effective collaboration across stakeholder groups: the shipping companies moving beyond cost-driven decisions toward sustainability-focused approaches, the recycling facilities fast-tracking compliance upgrades and capacity expansions, regional and National Governments providing regulatory clarity and financial support, and third-party organizations expanding verification and due-diligence capabilities.

The solutions pathways demand immediate action in the form of accelerated capacity expansion through both facility upgrades and new developments, advanced technology & automation adoption to improve efficiency along with environmental performance, innovative financing mechanisms to reduce investment roadblocks, and enhanced international cooperation to coordinate global capacity building.

The window for proactive response is rapidly narrowing. The projected surge in recycling demand will begin materializing within this decade. Delay in addressing capacity constraints risks creating bottlenecks that could force compromise of environmental and safety standards, undermining decades of progress toward responsible recycling practices.

The maritime industry stands at a defining moment where proactive leadership can transform a looming crisis into an opportunity for sustainable industry evolution. The challenge is formidable, but the consequences of inaction far exceed the costs of coordinated response. The time for action is now.

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